

Statistics Lecture Notes

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Introduction

These are the lecture notes for Math 146 (34644) at Seattle Central College. These notes will be updated throughout the semester. Please note that these are built on top of your assigned readings and it is recommended that you go through those thoroughly first!

Lecture numbers will not exactly correspond to the numbers in the navigation. Here I will keep a running list on what is covered each day.

1. Wed, Sept 30: Went through **Exploring Data** notes.
2. Mon, Oct 5: Went through **Summary Statistics** notes. Also worked on **Topic 1 Check In** (worksheet available with solutions on WAMAP under Additional Class Resources). See additional practice problems at the end of **Summary Statistics** for more practice.

1 Exploring Data

1.1 Overview

In these notes we will look at a few common methods for visualizing data. There are two main types of data we will be concerned with in this class: **categorical** and **quantitative** data. Categorical data is when the results are descriptive and can be organized into a finite number of answers (or categories). Examples of categorical data include: favorite color, blood type, methods of transportation, favorite food, etc. Quantitative data is data that is numerical *and the numbers are representative of actual amounts*. This can include: weights, heights, times, exam scores, etc.

Example 1.1. Discuss with a partner whether the following are **categorical** or **quantitative**:

- Age
- Birth month
- Birth year
- Hair color
- Zip code
- Number of siblings
- Jersey numbers (in sports)
- Satisfaction score from 1-5

The tools we use to visualize data depend on whether the data is categorical or quantitative.

1.2 Bar and Pie Charts

Bar and **Pie charts** are two ways to visualize the distribution of **categorical** data. Suppose 250 individuals were polled about how they commute to work and the following results were obtained,

Category	Count
Car	134
Public Transit	58

Category	Count
Bike	44
Walk	24

We can visualize this data using either a bar or pie chart.

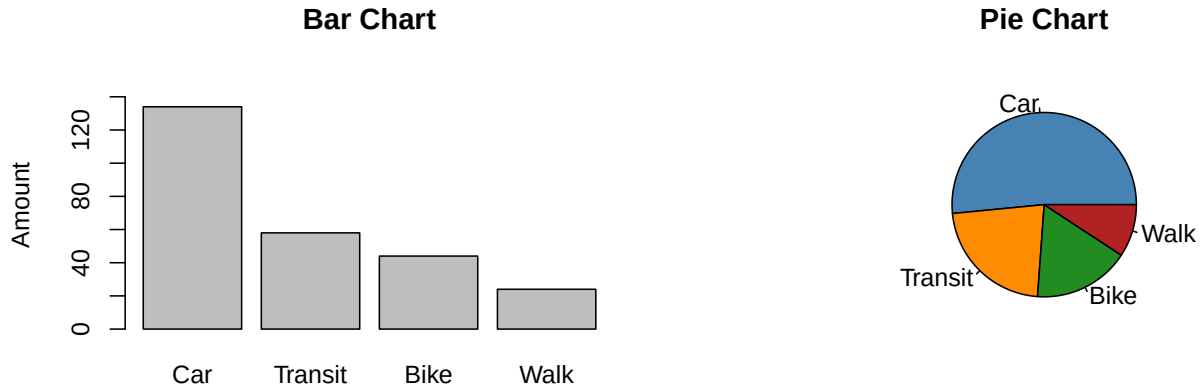


Figure 1.1: The same data shown as a bar chart (left) and a pie chart (right).

Creating a bar chart is simple, each **categorical variable** gets a bar where the height is represented by the number of respondents in the dataset. To create a pie chart, each category gets a slice of the pie proportional to its percent of the responses.

Category	Count	Percentage
Car	134	53.6%
Public Transit	58	23.2%
Bike	44	17.6%
Walk	24	9.6%

Example 1.2. A class of 32 students was surveyed about whether they prefer morning, afternoon, or evening classes. The results are given in the table below.

Time of Day	Count
Morning	10
Afternoon	14
Evening	8

To compute the percentage of votes of each category, we divide the quantity by the total number of students, 32.

Time of Day	Count	Percentage
Morning	10	$10/32 = 31.25\%$
Afternoon	14	$14/32 = 43.75\%$
Evening	8	$8/32 = 25\%$

Pie Chart

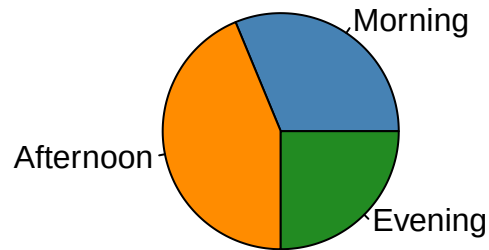


Figure 1.2: A pie chart depicting student class time preferences.

! Picking the Best Visual

In what ways are bar and pie charts similar and in what ways are they different? What are some benefits to a pie chart over a bar chart (and vice versa)?

💡 Answer

A pie chart is better when the main question is about part of a whole, like whether one category is more than half the total.

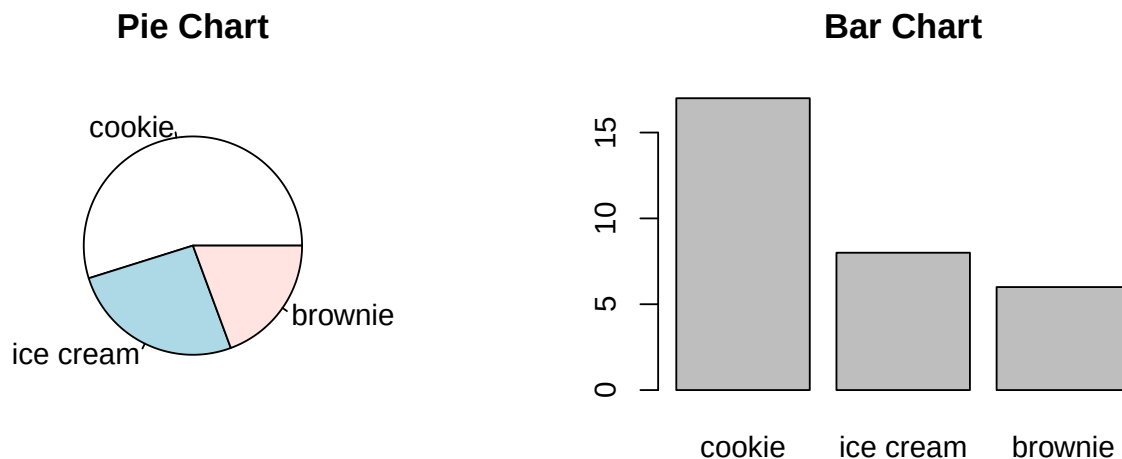


Figure 1.3: Bar and pie charts showing respondents favorite food.

A bar chart is better when we are comparing many close values. For example consider the favorite food of 100 respondents,

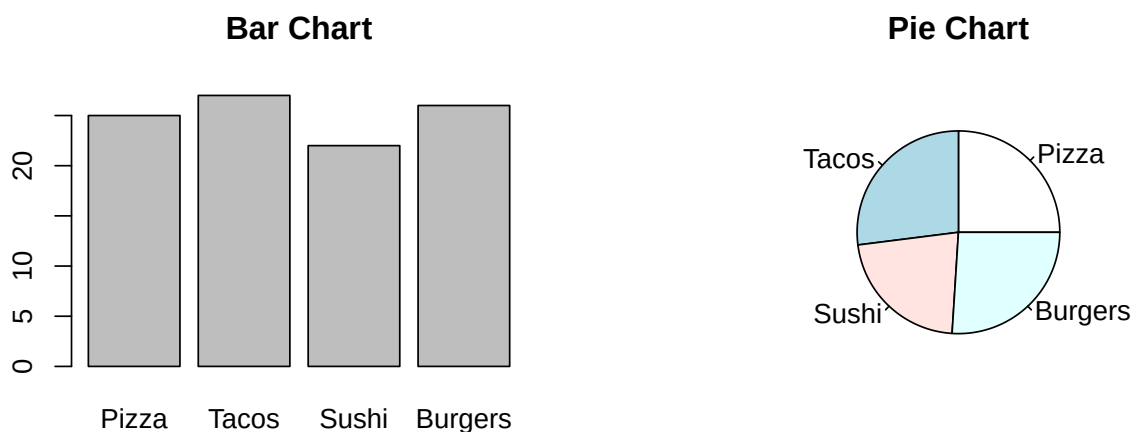


Figure 1.4: Bar and pie charts showing respondents favorite food.

Don't worry so much about creating a pie or bar chart by hand as we will use technology in this class. The below example explain how to create the graphs using StatCrunch.

Example 1.3 (Basic Visuals in StatCrunch). A company has collected data on 500 customers. [Open this dataset in StatCrunch](#). Use StatCrunch to create a bar chart of customer satisfaction by doing the following:

1. Under Graph, select Bar Plot, then With Data.
2. In the pop up screen, select the column SatisfactionRating. Set Type to Frequency and then hit Compute!

Next create a pie chart of the customer regions by selecting Graph > Pie Chart > With Data.

1.3 Misleading Visuals

Some visualizations can be misleading! Sometimes this is just poor judgement, but sometimes this can be done to make a specific point.

Example 1.4 (Bad Visualizations). Can you determine what makes these visuals misleading? Do you think it was done to make a specific point?

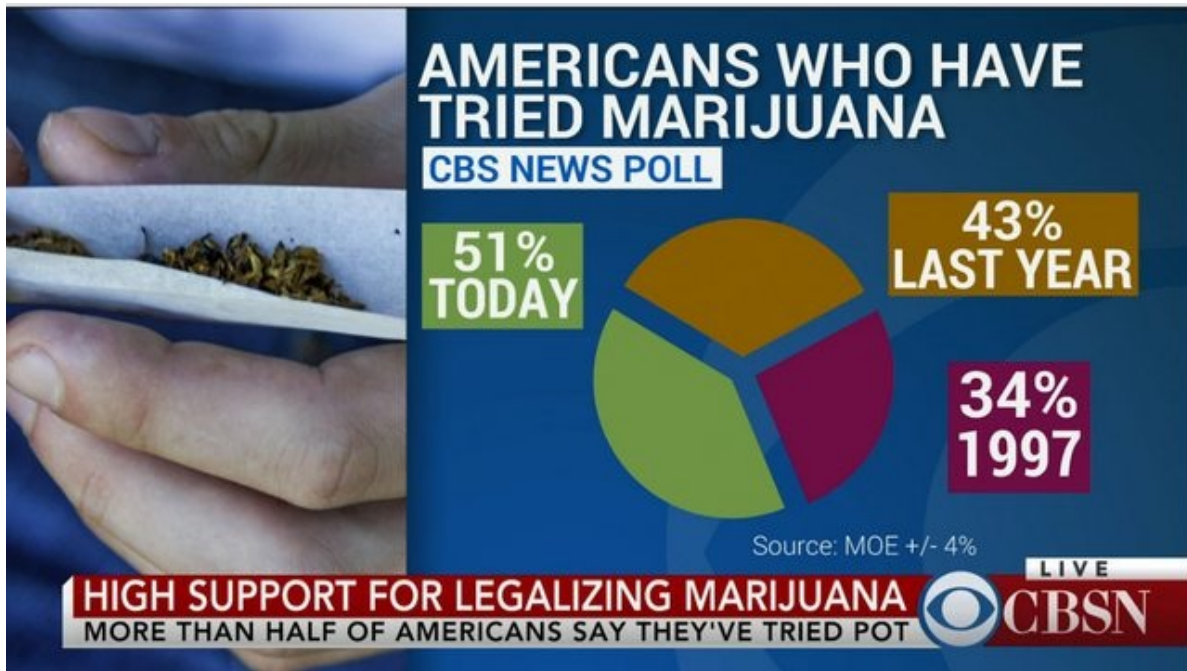


Figure 1.5: Image from www.codeconquest.com/blog/12-bad-data-visualization-examples-explained/

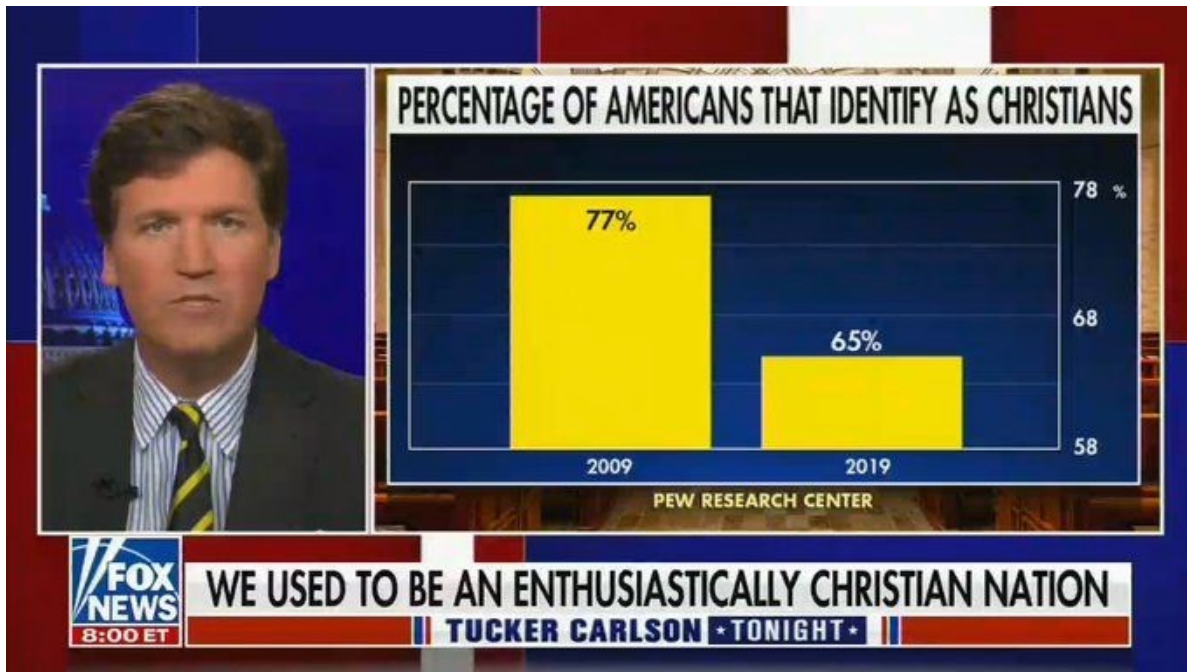


Figure 1.6: Image from www.codeconquest.com/blog/12-bad-data-visualization-examples-explained/

1.4 Histograms

Bar and pie charts are a great way to visualize *categorical* data. Now we want to address ways to visualize **quantitative** data, or data that is in the form of numbers.

Example 1.5 (Quantitative Data). Some example of quantitative data include: test scores, age, height, weight, (systolic) blood pressure, etc.

A good way to look at quantitative data is to use a **histogram**. A histogram breaks the range of all possible values into smaller intervals, then counts how many observation fall into each window. This looks kind of like a bar chart, but the bars sit right next to each other.

Example 1.6. Students in a calculus class obtain the following scores on the midterm exam,

100, 85, 91, 76, 88, 82, 95, 62, 84, 79, 90, 73, 87, 81, 92, 77, 86, 83, 80, 89, 74, 93, 85, 79, 82

The lowest score is a 62 and the highest score is 100. Let's break down the range $[60, 100]$ into intervals of length 5.

Interval	Count
(60,65]	1
(65,70]	0
(70,75]	2
(75,80]	5
(80,85]	7
(85,90]	5
(90,95]	4
(95,100]	1

From this we can make a histogram.

Distribution of Exam Scores

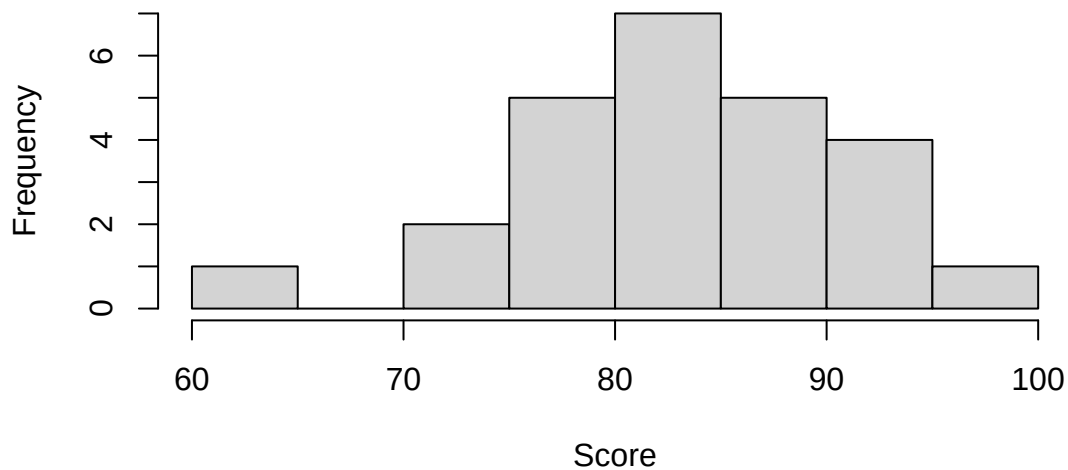


Figure 1.7: A histogram of test scores.

It is typical to call the intervals of a histogram **bins**, although the reading on WAMAP will call them **classes**. A histogram should always have bins of the same length. It can be misleading to use different bin lengths in the same way it is misleading to use breaks on the vertical axis of bar charts.

! Histogram Versus Bar Chart

Histogram “bars” touch, while bar chart bars do not. Why do you think this distinction is made?

💡 Answer

Histogram bars touch because the bins cover a continuous range of numbers with no gaps between them. Bar chart bars are separated because each bar is a distinct category, and there's nothing connecting the categories together.

Most computer programs will set automatic bin sizes based on the distribution of the data you get it, although you will typically have an option to manually input bin specifications. In StatCrunch you are given the optional inputs of the starting value and the bin width.

! Bin Endpoints

For whatever computer program you use, make sure you know how the **endpoints** of your bin intervals are treated. In Example 1.6, I wrote intervals like $(70,75]$ which means 70 is excluded from the bin and 75 is included in the bin. The statistical programming language R (what generates the graphs in these notes) uses right inclusive bins, however StatCrunch and Google Sheets use *left* inclusive bins.

The histograms below represent *the same* data, just with different bin sizes.

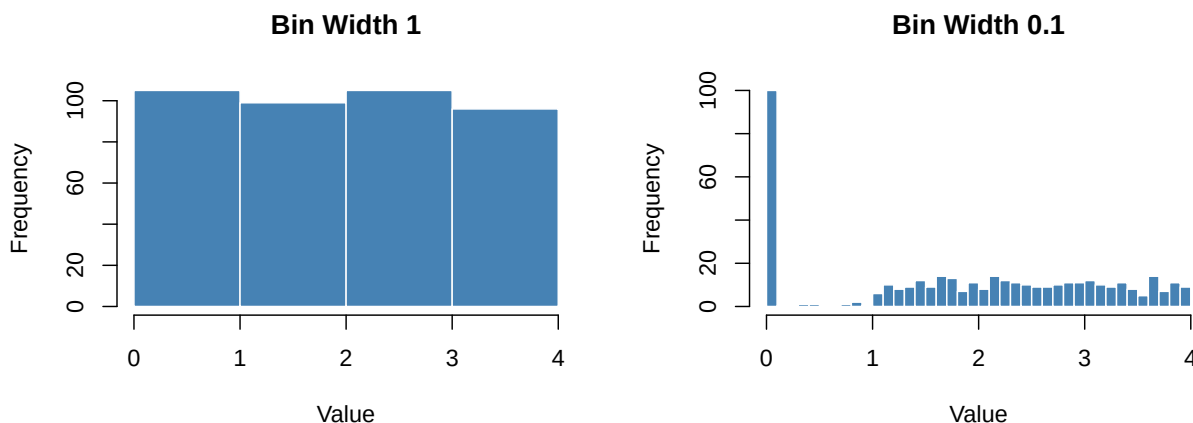


Figure 1.8: Histograms of the same dataset with different bin widths.

Example 1.7. Discuss the following with a partner or two:

- Which histogram do you think is better for the above data set and why?
- What goes wrong if bins are too wide? What if bins are too narrow?
- Is there one “correct” bin width?
- Could someone manipulate bin size to mislead the public about a dataset?

Example 1.8 (Histograms on StatCrunch). [Reopen the customer dataset in StatCrunch.](#) Create a histogram of customer ages by going to Graph > Histogram.

Example 1.9 (High Tide Heights). The following histogram shows 60 high tide heights over the month of July at a beach. What **percentage** of high tides were more than 2 meters?

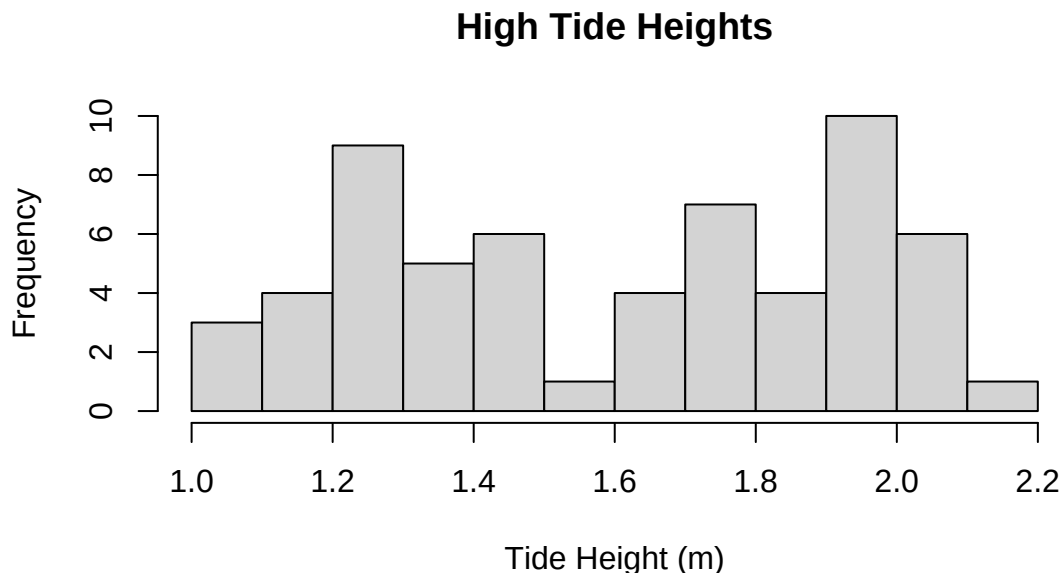


Figure 1.9: Histograms of high tide heights in July.

Answer

We want to compute,

$$\frac{\text{high tides over 2 m}}{\text{total number of high tides}} \cdot 100,$$

where multiply by 100 turns the decimal into a percent. The total number of high tides is 60 as stated in the problem. If not given explicitly, we could carefully sum the quantity in each bin.

The number of high tides that were more than 2 meters is given by the total in the last two bins, $6 + 1 = 7$.

$$\frac{7}{60} \cdot 100 \approx 11.67\%$$

! Information Lost

What kinds of information do we lose by only seeing a histogram of the data?

We can use words to describe the shape of distributions. Some common terms are:

- A distribution is **symmetric** if the left and right sides are roughly mirror images of each

other.

- A distribution is **skewed right** if it has a long tail stretching to the right. Most values are low, with a few unusually high values.
- A distribution is **skewed left** if it has a long tail stretching to the left. Most values are high, with a few unusually low values.

The direction of the skew is the direction of the *tail*, not the direction of the peak. A skewed-right distribution has its peak on the left.

We also describe distributions by their number of peaks. A distribution with one peak is **unimodal**, and a distribution with two clear peaks is **bimodal**. The symmetric and skewed examples below are all unimodal.

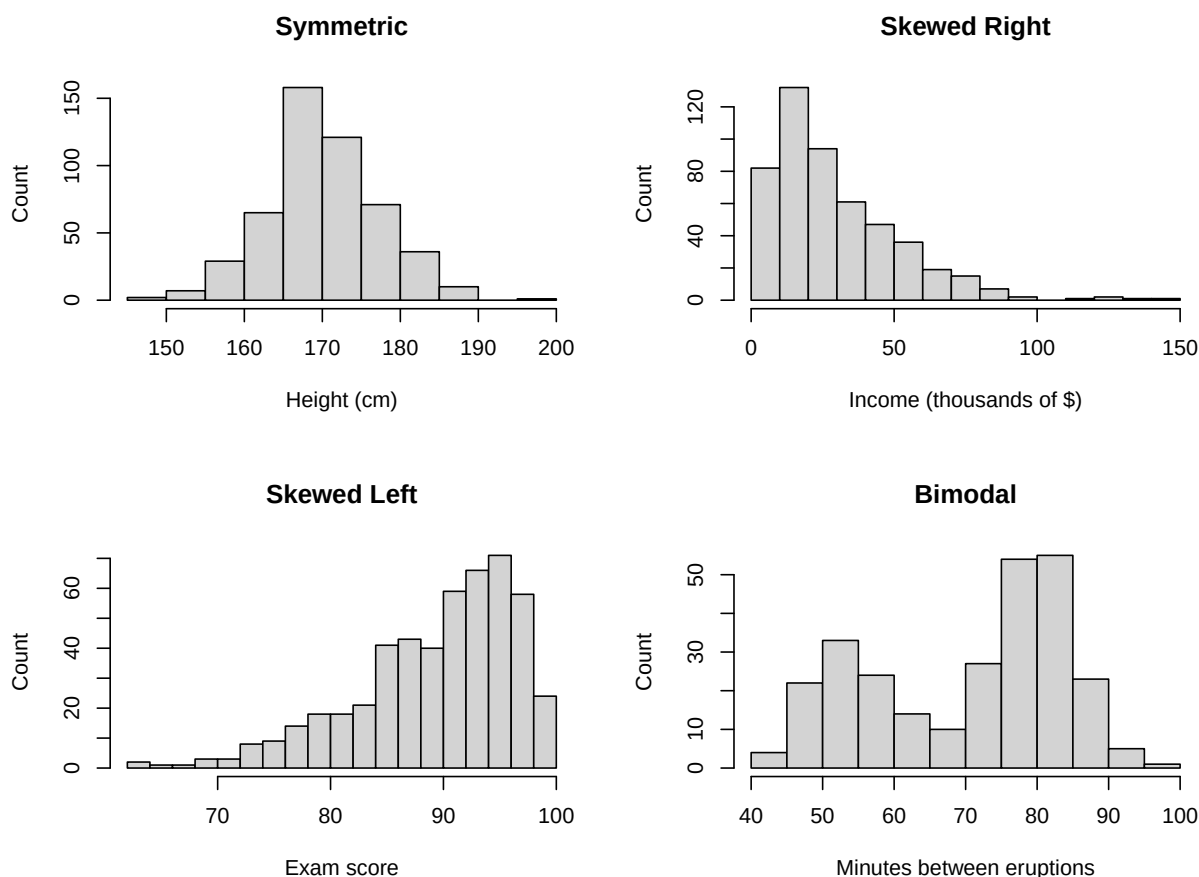


Figure 1.10: Four common distribution shapes.

Shape	Real World Examples
Symmetric	Adult heights (for one sex), repeated measurements of the same quantity, standardized test scores
Skewed right	Household income, home prices, number of social media followers, time spent waiting in line
Skewed left	Scores on an easy exam, age at death in countries with high life expectancy
Bimodal	Customers in a restaurant, shoe sizes of a population not separated by gender

! Why Bimodal?

A bimodal distribution often means the data contains two different groups mixed together. For the restaurant example, what are the two groups? Can you think of another variable that might be bimodal?

Example 1.10 (Plant Heights). A biology class grew 40 bean plants under identical conditions and measured each plant's height, in centimeters, after six weeks. The histogram below shows the results.

Histogram of heights

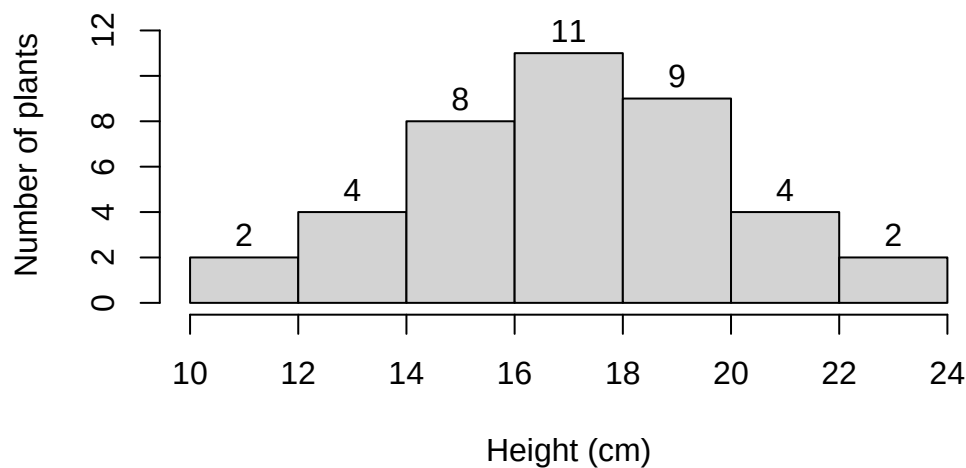


Figure 1.11: Histogram of bean plant heights after six weeks.

1. What is the bin width of this histogram?

2. Describe the shape of the distribution. Around what height is the center?
3. What percentage of plants grew taller than 20 cm?
4. What percentage of plants had heights between 14 and 20 cm?
5. Can you tell from the histogram how tall the shortest plant is?

 Answer

1. Each bin covers 2 cm, for example 16 to 18 cm.
2. The distribution is roughly **symmetric**: the tallest bar is in the middle, at 16–18 cm, and the bars decrease at about the same rate on both sides. The center is around 17 cm.
3. The last two bins contain plants taller than 20 cm: $4 + 2 = 6$ plants.

$$\frac{6}{40} \cdot 100 = 15\%$$

4. The bins from 14 to 20 cm contain $8 + 11 + 9 = 28$ plants.

$$\frac{28}{40} \cdot 100 = 70\%$$

5. No. We only know the shortest plant is somewhere between 10 and 12 cm. The histogram does not show individual values.

2 Summary Statistics

We've given a few examples of ways to visualize data, now we want to talk about some important **summary statistics**. Summary statistics are quantitative values one can use help describe the shape of a distribution. We will talk about measures of **center** and measures of **spread**.

Almost always, summary statistics can only be used on *quantitative data*.

2.1 Measures of Center

The goal of measures of *center* is to describe the value of a typical datapoint. There are three measures of center we will cover:

- **Mode:** The value(s) with the highest frequency in the dataset. Mode is the *only* measure of center that can be used with categorical data.
- **Mean:** The average of the values. If a dataset has n values $x_1, x_2, \dots, x_n$, then the mean $\bar{x}$ is given by

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}.$$

- **Median:** The midpoint of the distribution. It is the number such that half the observations fall above and half fall below.

Example 2.1 (Classroom Ages). A college class has 22 students with the following ages: 18, 22, 21, 21, 21, 18, 20, 42, 23, 44, 18, 18, 21, 18, 22, 18, 22, 23, 20, 38, 20, 34. Compute the mean, median, and mode.

[See this data on StatCrunch to perform the computations there.](#)



Answer

To compute the mode and median it helps to order the data from smallest to largest.

18, 18, 18, 18, 18, 18, 20, 20, 20, 21, 21, 21, 21, 22, 22, 22, 23, 23, 34, 38, 42, 44

The mode is 18, which occurs in the dataset 6 times.

Since the dataset is 22 students, the median is the value in-between the 11th and 12th observations. The 11th and 12th observation both happen to be 21, therefore the median is 21.

The mean is,

$$\frac{6 \cdot 18 + 3 \cdot 20 + 4 \cdot 21 + 3 \cdot 22 + 2 \cdot 23 + 34 + 42 + 44}{22} = \frac{522}{22} = 23.73.$$

Notice that the mean is slightly larger than the median this is because there are a handful of older students with *outlier* ages. The mean gets pulled by outliers more than the median does.

! Median on Datasets of Even Size

For this course, to compute the median when the dataset has an *even* number of observations average the two observation in the middle. This is a typical convention chosen in statistics.

The difference between the median and mean of a distribution can reveal information about the shape of the distribution. In particular, if a distribution is **skewed** or **symmetric**.

Household Income Distribution

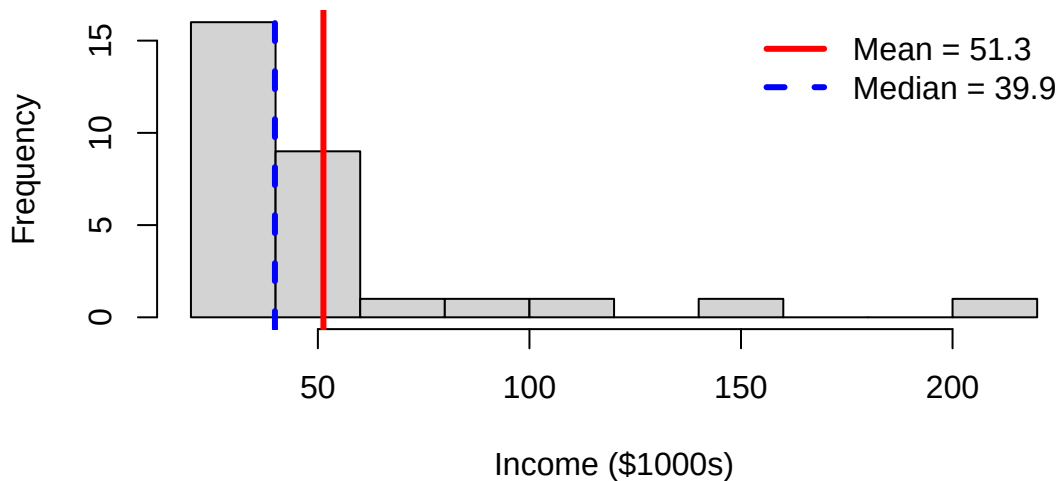


Figure 2.1: Right-skewed household income distribution.

When a distribution is skewed, the mean is pulled more towards the **tail** than the median.

Soccer Team Player Heights

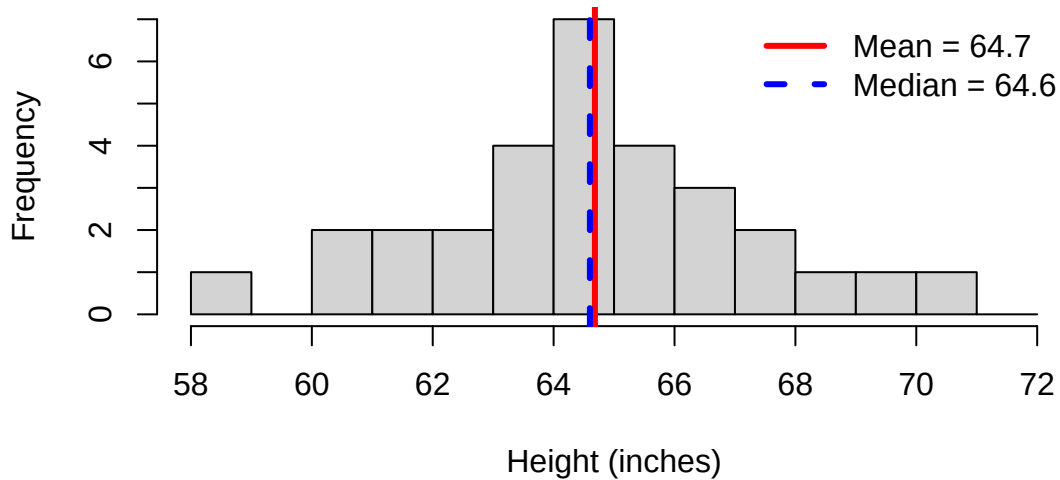


Figure 2.2: Approximately symmetric distribution of soccer team heights.

! Mean Versus Median

The reason **outliers pull the mean more than the median** is because mean minimizes the error *squared*, while the median minimizes the absolute error. Squaring the error means outliers will have extra impact. This fact is outside of the scope of this course, but I am happy to chat about it for curious minds!

Example 2.2 (Creating Data). Using StatCrunch complete the following:

1. Create a dataset of 10 student ages where the median is 20 and at least one student is not 20.
2. Is it possible to do this and have no student be 20 years old?
3. If we increased the size of the dataset to 15 students can you still create a dataset with median 20 *and* no students are 20 years old?

💡 Answer

1. One possible answer: 18, 18, 18, 19, 20, 20, 20, 21, 22, 23.
2. Yes it is! Since the dataset has an *even* size, we can make it so the 5th entry is 19 and the 6th entry is 21. Such as: 18, 18, 18, 19, 19, 21, 22, 23, 25.
3. This is **not** possible if the dataset has an *odd* size. When there are 15 students, the

age of the 8th youngest student is the median—so that student must be 20.

Example 2.3.

1. Using StatCrunch or Google Sheets, create a dataset of 25 non-negative integers (e.g. whole numbers 0 and larger) where the mean is less than the median. Plot the histogram of the data.
2. (Challenge) If I have a dataset of 25 **non-negative integers** and the median is 50, what is the smallest possible mean?

Answer

1. Your dataset should be **skewed to the left** in this example.
2. The smallest possible mean occurs when 12 data points have value 0 and 13 have the value 50. This guarantees the median is 50. So the mean is

$$\frac{12 \cdot 0 + 13 \cdot 50}{25} = 26.$$

We can't make any of numbers smaller or else we would either change the median or break the non-negative rule. Making any of the numbers bigger would increase the mean!

When to use mode?

Discuss: what might be a situation where the mode is the most relevant summary statistic to use?

Tip

Mode is the only measure of center that can be used for categorical data. Also, mode is great for bimodal (or multimodal) distributions!

Nobody's Average

In 1950 the airforce measured 140 physical dimension of 4063 pilots and designed the cockpit to match the average (mean) dimensions. Lieutenant Gilbert S. Daniels then compared this “average pilot” to individual pilot records to show that no pilot was within 15% of the average on the ten most relevant dimensions. In other words, the cockpits were designed to fit no one! This led to the adaptation of adjustable seats.

2.2 Measures of Spread

Example 2.4 (Commute Times). [Data on StatCrunch](#). The below histograms depict the commute time of 500 people who work in Town A and Town B. How are the datasets similar and how are they different?

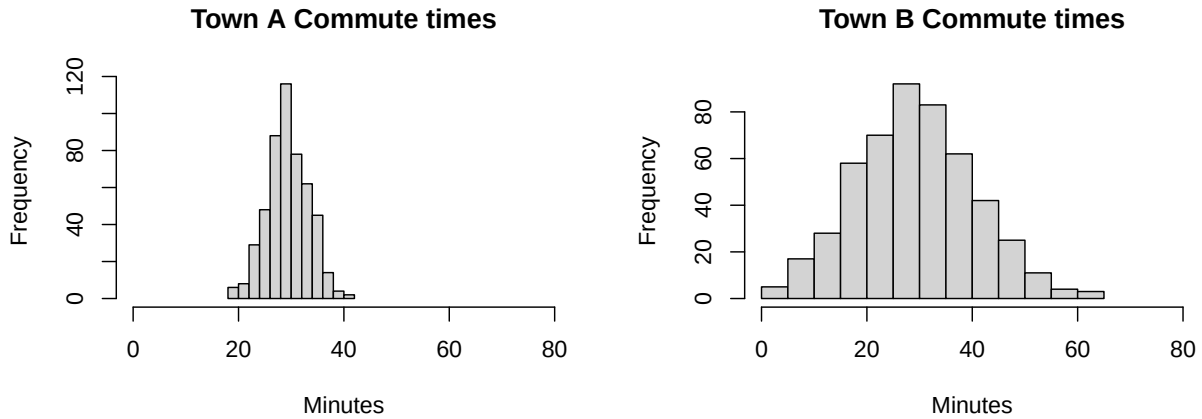


Figure 2.3: Commute times in two different towns.

The above example illustrates that it is important to have statistics that describe the *spread* of a distribution. There are a few we will investigate today.

2.2.1 Range

The **range** of a dataset is the largest observation minus the smallest observation.

Example 2.5. [The commute times data is given here](#). Compute the range of the commuting data for each town.

Answer

For Town A, the longest and shortest commute times are 42 and 18 minutes, respectively. So the range for Town A is $42 - 18 = 24$.

For Town B, the longest and shortest commute times are 62 and 1 minute, respectively. So the range for Town B is $62 - 1 = 61$.

Range does give us an initial look at the spread of the distribution, but **outliers** can obfuscate how range relates to spread. Consider two classes with the following histograms of exam scores:

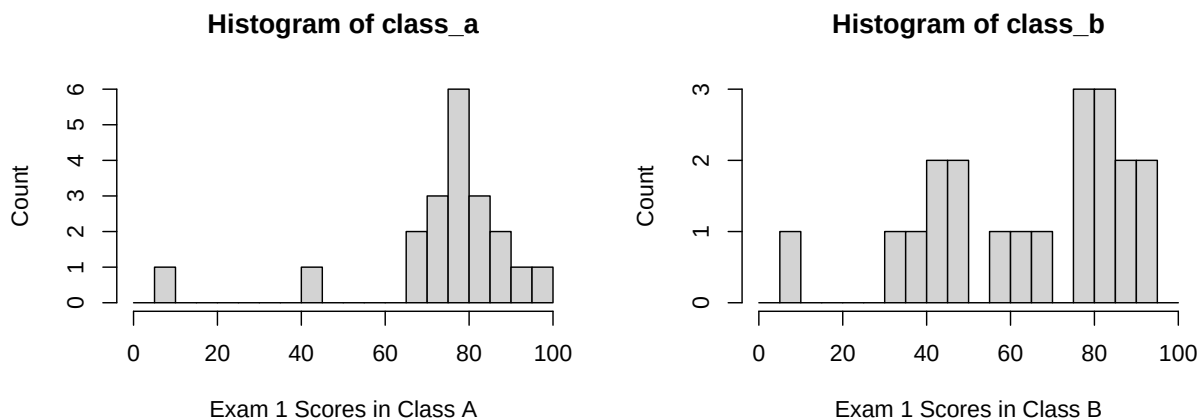


Figure 2.4: Scores on Exam 1 in Class A and Class B

The range of both distributions is (approximately) the same, but the spread looks very different. Only 2 out of 20 students scored below a 60 in Class A, while 8 students (out of 20) scored below a 60 in Class B.

2.2.2 Inter-Quartile Range (IQR)

Recall that the **median** divides the observations into two parts of equal size. This is because we defined the median as the value which half the observations are below and half are above.

Quartiles divide the data into 4 equal parts. Specifically we define:

- The first quartile (or lower quartile), Q_1 , is the value such that 25% of the data lies below this value.
- The second quartile (or median), M , is value where 50% of the data lies below this value.
- The third quartile (or upper quartile), Q_3 , is the value such that 75% of the data lies below this value.

Computing the min, Q_1 , Q_2 , Q_3 , and max of a data set is known as the **five-number summary**.

! Important

Note that Q_1 and Q_3 are computed differently on Google Sheets than on StatCrunch (and in this class). While they are effectively the same, it will sometimes cause incorrect homework answers.

Example 2.6. Below are the exact list of Exam 1 scores for Class A and Class B (from the histograms in Figure Figure 2.4). Compute the five-number summary for each class. What differences do you observe about the distribution from the statistics?

Class A	Class B	Position Number
10	10	1
44	34	2
66	36	3
70	44	4
71	45	5
73	48	6
74	50	7
76	56	8
77	64	9
79	66	10
80	76	11
80	77	12
80	80	13
82	82	14
83	83	15
83	84	16
86	88	17
87	89	18
91	91	19
96	95	20

Answer

Each class list is already ordered for us, so we have:

- Min = 1st grade $\rightarrow$ Class A = 10, Class B = 10
- Q1 = between 5th and 6th grade $\rightarrow$ Class A = 72, Class B = 46.5
- M = between 10th and 11th grade $\rightarrow$ Class A = 79.5, Class B = 71
- Q3 = between 15th and 16th grade $\rightarrow$ Class A = 83, Class B = 83.5
- Max = 20th grade $\rightarrow$ Class A = 96, Class B = 95

The **inter-quartile range** (IQR) is defined as $Q3 - Q1$.

Example 2.7. Compute the IQR for the exam scores by class.

Answer

For Class A, $IQR = Q3 - Q1 = 83 - 72 = 11$. For Class B, $IQR = 83.5 - 46.5 = 47$.

! IQR Versus Range

Why might IQR be seen as a better gauge of spread than range?

2.2.3 Outliers

An **outlier** is an observation that lies far away from the general distribution. The readings define outliers as points that are:

- Smaller than $Q1 - 1.5 \cdot (IQR)$, or
- Larger than $Q3 + 1.5 \cdot (IQR)$.

Do not think of this as a set in stone rule. In fact, the multiplier 1.5 was not chosen with any real mathematical theory behind it. Use these values to flag *potential* outliers in the dataset, but context is an important factor when determining what is an outlier. Moreover, **do not** automatically remove outliers from a dataset

Example 2.8 (Bimodal Commute Times). We have the following list of commute times for a neighborhood:

24	25	26	27	27	28
28	29	29	30	30	30
31	31	32	32	33	33
34	34	35	35	36	36
37	38	39	40	41	42
72	74	76	78	80	82

Let's start by looking at the five-number summary.

[1] 24.0 29.5 33.5 39.5 82.0

From this we can compute the lower and upper fence:

$$\text{lower} = 29.5 - 1.5(10) = 14.5, \quad \text{upper} = 39.5 + 15 = 54.5$$

That means 6 data points are flagged as outliers using this rule: 72, 74, 76, 78, 80, 82. Let's look at the histogram of the distribution,

What we see from the histogram is a **bimodal** distribution. There is one subset of the neighborhood that commutes a shorter distance (think across town), while a small subset of the neighborhood commutes to a different city. This is an example where the $1.5 \times IQR$ rule is a bad fit.

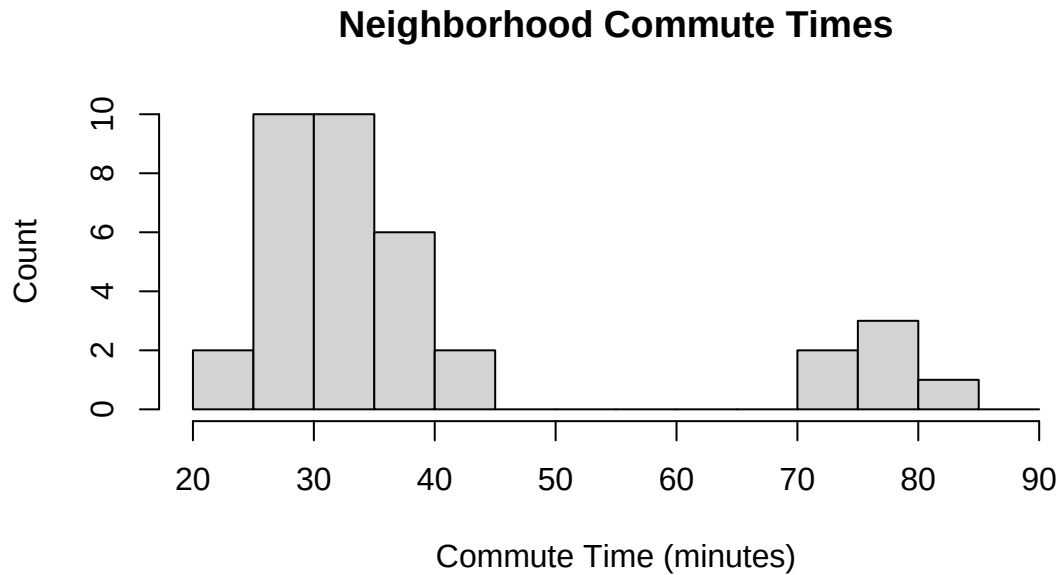


Figure 2.5: Histogram of neighborhood commute times.

! Removing Data

There is a false notion that outliers should be removed from the data set but this is **not true!** Outliers should be expected in data and are part of the underlying distribution. Only remove data points if they are the result of mistake in data entry or collection, or if they can be explained by a fundamentally different process which you are not interested in investigating.

2.3 Boxplots

Boxplots are a way to visualize the five number summary as well as outliers. Let's see this with an example.

Example 2.9 (Boxplots in StatCrunch). [For this example use this data on StatCrunch.](#) To make a boxplot of the data select Graph > Boxplot. Just select annual_income and then compute.

The box of the boxplot goes from Q1 to Q3 and the median is shown as a horizontal line. The two sticks (called whiskers) on either side of the box go until the largest data point *that isn't an outlier* and the stars (called fences) indicate outlier values. If we deselect the box indicating "Use fences to identify outliers" in StatCrunch then the whiskers will go all the way until the end of the range.

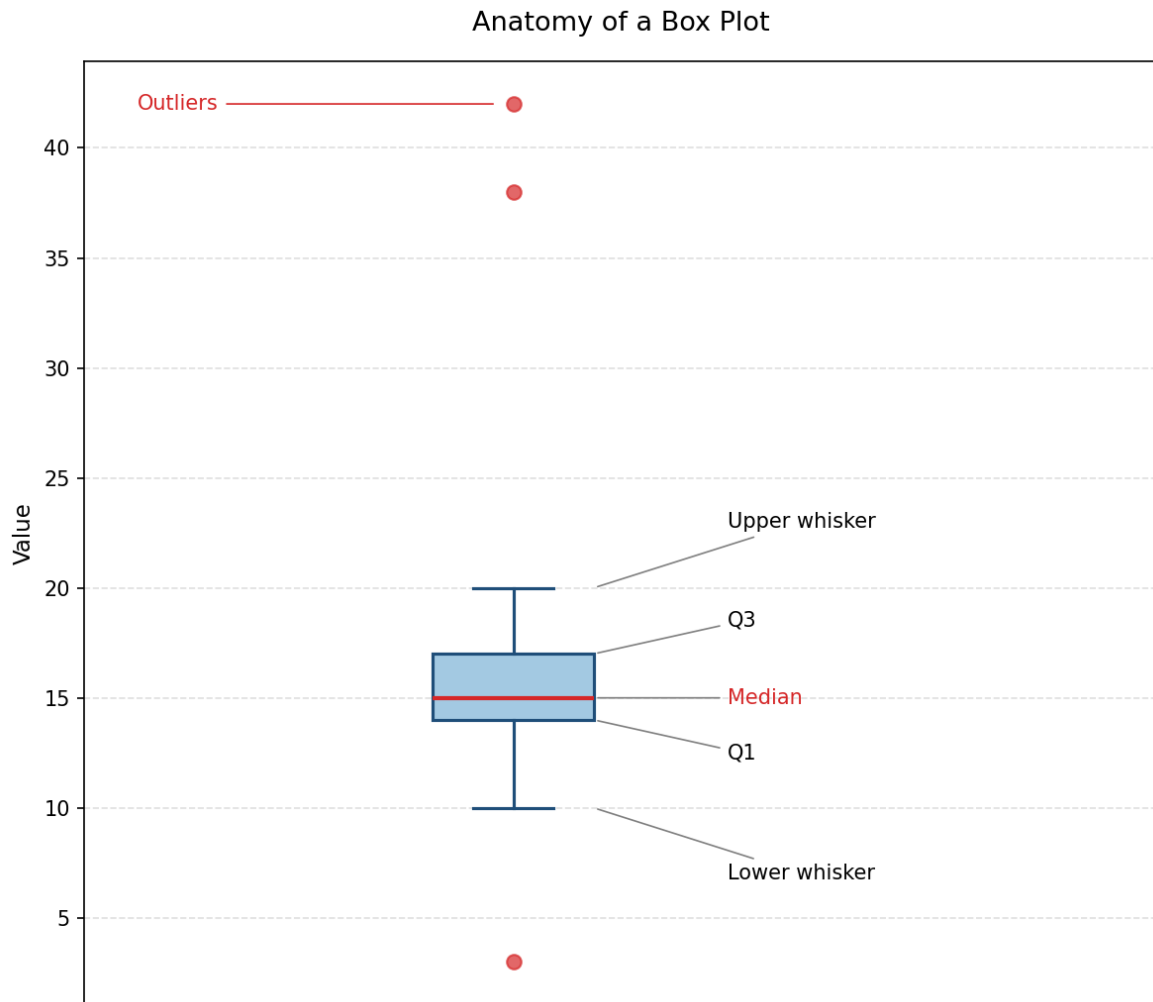


Figure 2.6: Labeled boxplot.

Example 2.10. The boxplots below show the typed words per minutes for support agents and software engineers. Compare the two distributions using what you know about boxplots.

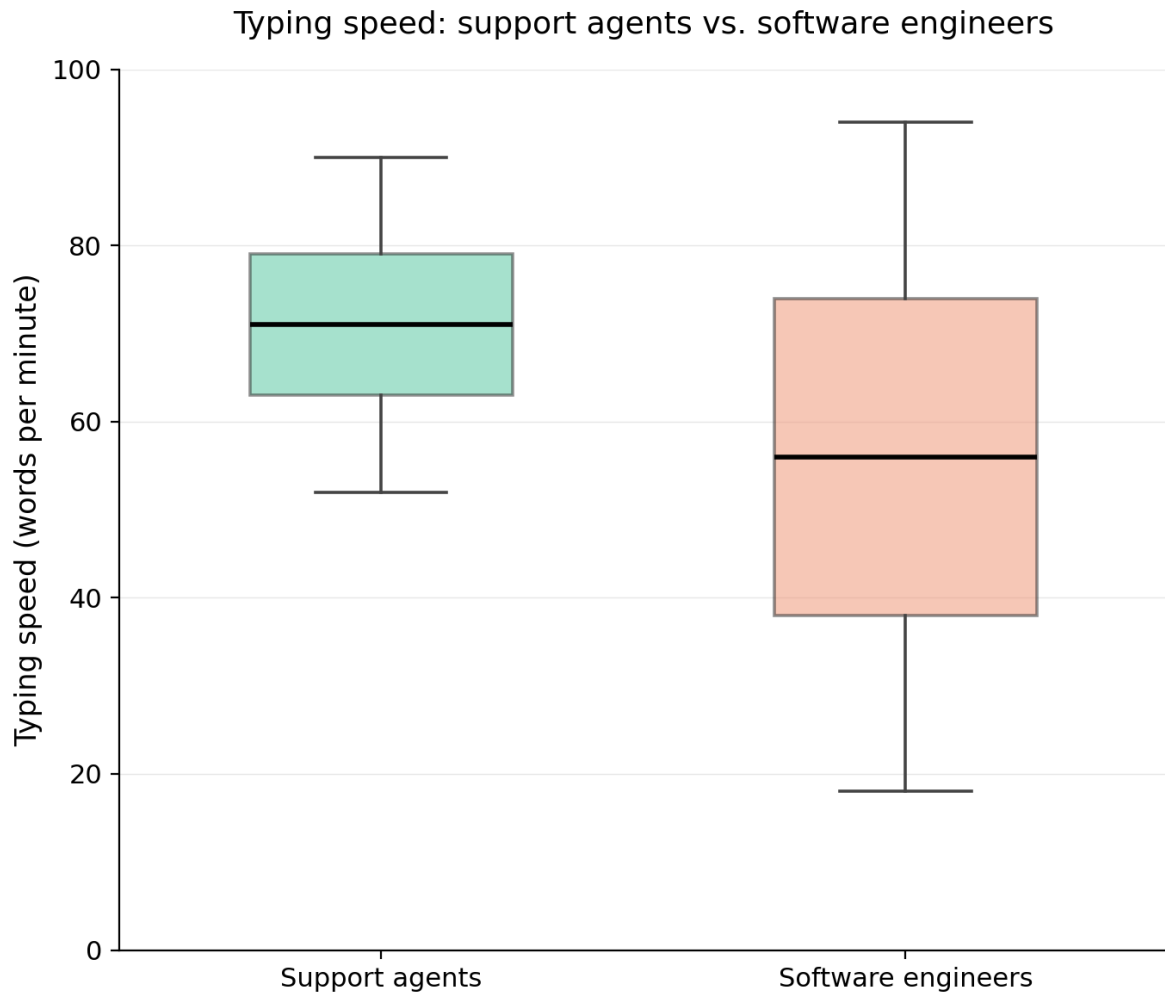


Figure 2.7: Typing speed boxplots for support agents and software engineers.

💡 Answer

The median support agent is a faster typer than the median software engineer. Software engineers have a much larger *variation* in typing speeds than support agents.

2.4 Standard Deviation

Here we will define and use the **sample standard deviation**. This is slightly different, but related to, the *population* standard deviation as well as the variance. These are terms we will

define explicitly later. Given data points $x_1, \dots, x_n$ the **sample standard deviation** is defined by the formula

$$SD = \sqrt{\frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \dots + (x_n - \bar{x})^2}{n - 1}}$$

where $\bar{x}$ denotes the mean of the dataset. Let's see this formula in action with a small example,

Example 2.11. Anna denotes how many Reese's Peanutbutter Cups her 10 friends eat at a party. She gets the following amounts:

3, 4, 3, 6, 6, 2, 1, 8, 5, 5.

Compute the mean and standard deviation of the dataset.

 Answer

Let's compute the mean,

$$\bar{x} = \frac{3 + 4 + 3 + 6 + 6 + 2 + 1 + 8 + 5 + 5}{10} = \frac{43}{10} = 4.3.$$

Next compute the differences,

-1.3, -0.3, -1.3, 1.7, 1.7, -2.3, -3.3, 3.7, 0.7, 0.7.

Square and sum,

$$1.69 + 0.09 + 1.69 + 2.89 + 2.89 + 5.29 + 10.89 + 13.69 + 0.49 + 0.49 = 40.1$$

Now divide by $n - 1 = 10 - 1 = 9$ and take the square root,

$$SD = \sqrt{\frac{40.1}{9}} = \sqrt{4.456} \approx 2.11.$$

We claim that standard deviation is telling us *something* about the spread of the distribution, but we can be more precise than that. The standard deviation tell us the average distance from an given datapoint to the mean. A larger standard deviation means the data is more spread out, while a smaller standard deviation tells us the data is compact.

Example 2.12 (Standard Deviation in StatCrunch). [For this example use this data on StatCrunch.](#) Compute the standard deviation of the income dataset.

Example 2.13. Which histogram has the smallest standard deviation? The largest?

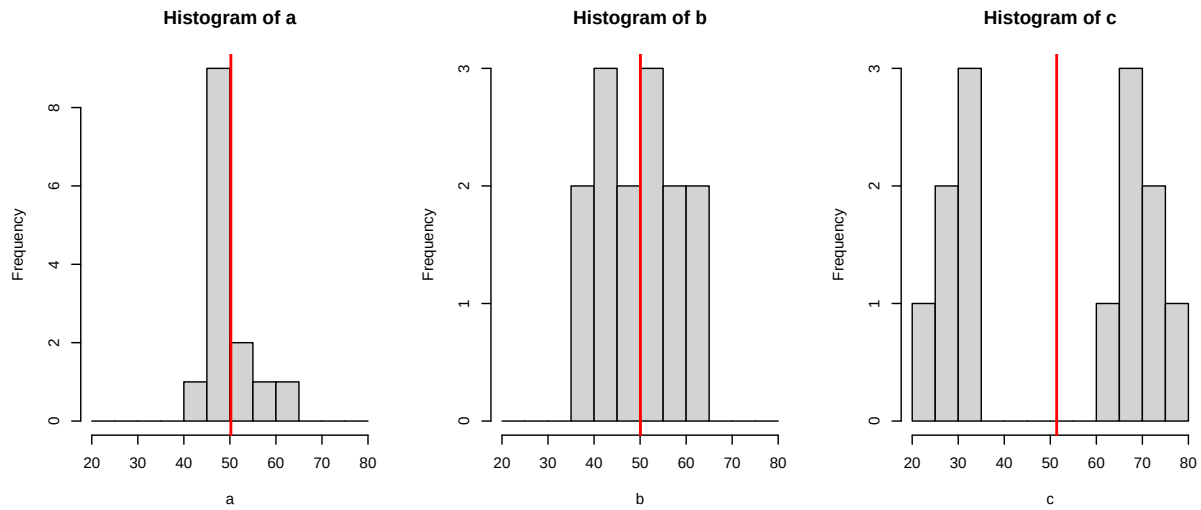


Figure 2.8: Histograms of data with similar mean and different standard deviations.

2.5 Normal Distribution and the 68-95-99.7

Quantitative data is **normally** distributed when its distribution looks like a symmetric “mound” or “bell” shape.

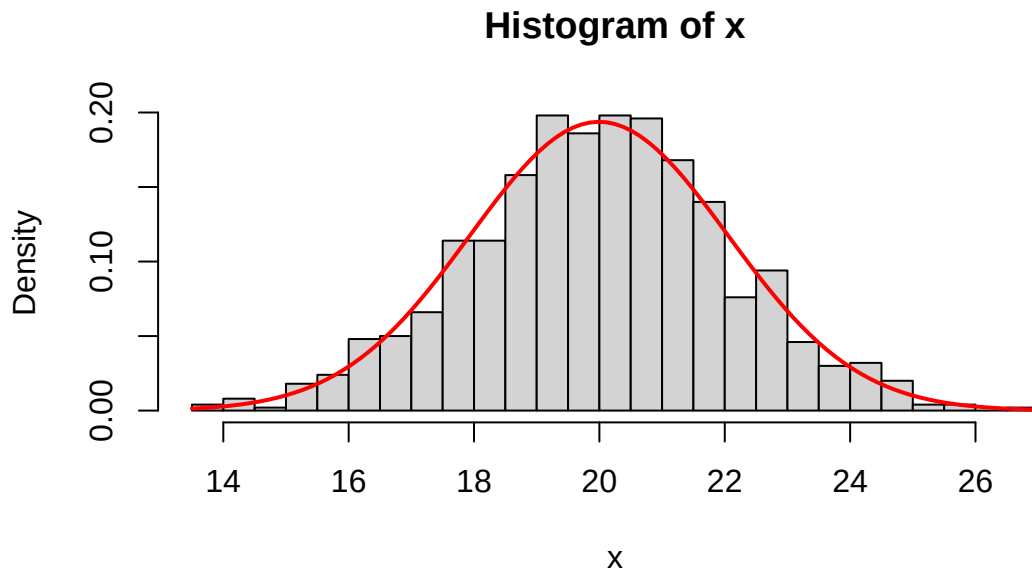


Figure 2.9: Histogram of approximately normal data.

Things that are approximately normal in distribution include:

- Baby birth weights
- Heights of adults by sex
- Blood pressure of adults
- Standardized testing scores

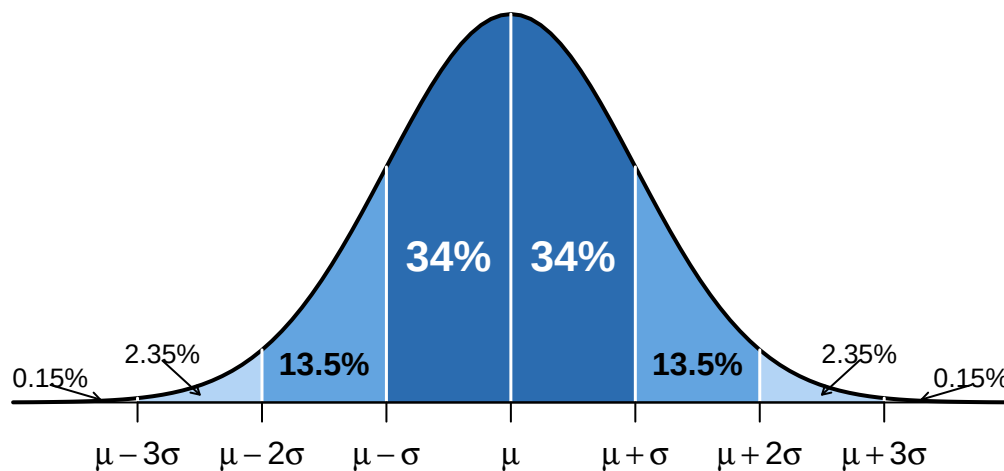
Statistics over time may have you thinking everything is normal, but this is not true! You should always question whether a distribution is normal or has some other distribution. Things that are *not* normal include:

- Income
- Time between earthquakes in a given region
- Number of shots taken in a soccer game
- Amount of time someone waits in a line

If a distribution is normal though, the **empirical rule** (or 68-95-99.7 rule) gives us a rough estimate of what percentage of the data falls within 1, 2, and 3 standard deviations of the mean. We have:

- 68% of the data falls within 1 standard deviation of the mean
- 95% of the data falls within 2 standard deviations of the mean
- 99.7% falls within 3 standard deviations of the mean

The Empirical Rule



Example 2.14 (Baby Weights). The mean birth weight at a hospital is 7.3 lbs with a standard deviation of 1.1 lbs. Assume that the birth weight is approximately normally distributed.

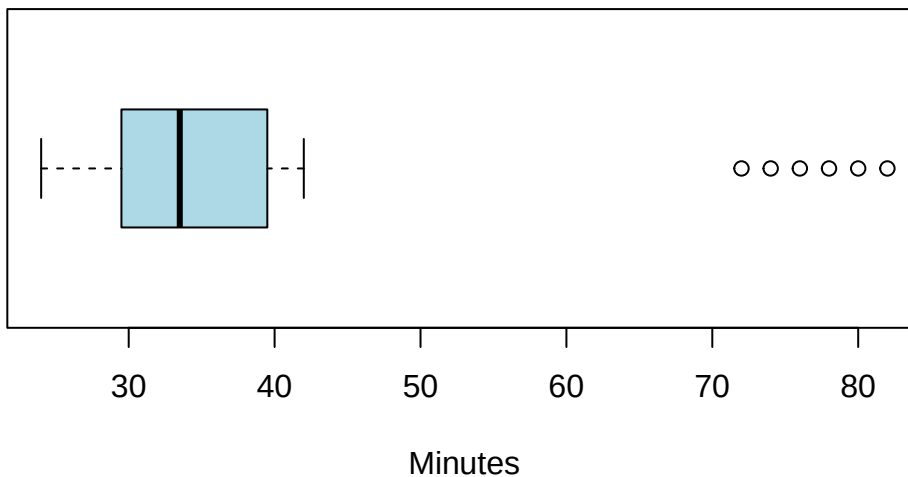
1. What percentage of babies weigh between 6.2 lbs and 8.4 lbs?
2. What percentage of babies weigh less than 4.0 lbs?

2.6 Extra Practice

1. Compute the mean and median for the annual income dataset. Based on these values is the data skewed to the right or left? Generate a histogram of the data to confirm your thoughts. [Data](#)
2. A survey recorded the time spent on homework in two different grades. The five-number summaries are recorded below. Compute the IQR for each grade. What are the main differences in the distributions?

	Grade 9	Grade 11
Minimum	60	40
Q1	90	70
Median	110	120
Q3	130	200
Maximum	160	260

3. The principal tells you that, based on the above summary data, 9th graders and 11th graders spend a similar amount of time on homework. How might you respond to the principal?
4. The box plot shows commute times (in minutes) for 36 employees.
 - a) Estimate the median, Q1, and Q3. What is the IQR?
 - b) Would the mean be larger or smaller than the median? Why?



5. A news article reports: “The mean commute time is 25 minutes with a standard deviation of 40 minutes.” What does this suggest about the shape of the distribution? Why might this statement be misleading?

6. Heights of adult women in the U.S. are approximately normal, with a mean of 64 inches and a standard deviation of 2.5 inches.
- a) What percentage of women are between 61.5 and 66.5 inches tall?
 - b) What percentage of women are taller than 69 inches?
 - c) What percentage of women are between 59 and 66.5 inches tall?

3 Examining Relationships

In Topic 1 we looked at ways to visualize data about a single variable. Now we want to learn ways to examine the relationship between *two variables*.

When we look at two variable we are interested in how one impacts the other. The **response variable** or dependent variable is the variable that gives the outcome of the study. The **explanatory variable** is the variable that claims to impact the response variable.

Example 3.1. Identify what is the explanatory and response variable in the following study questions:

1. Does the amount of fertilizer affect how tall tomato plants grow?
2. Does a car's weight impact its mileage?
3. Do people who sleep more have lower stress levels?

! Identifying Response and Explanatory

The roles aren't always clear-cut: sometimes either variable could reasonably influence the other. In these cases, the study design is key in dictating which is the explanatory and which is the response variable.

3.1 Side-by-Side Boxplots ($C \rightarrow Q$)

If the explanatory variable is categorical and the response variable is quantitative, then one way to visualize the potential differences in distribution is by using side-by-side boxplots.

Example 3.2. Below is a dataset of 189 births from a 1986 study at Baystate Medical Center (in Massachusetts). What similarities and differences do you notice between babies born from smokers versus nonsmokers? What can you not tell by the boxplots that you might be interested in knowing about the data?

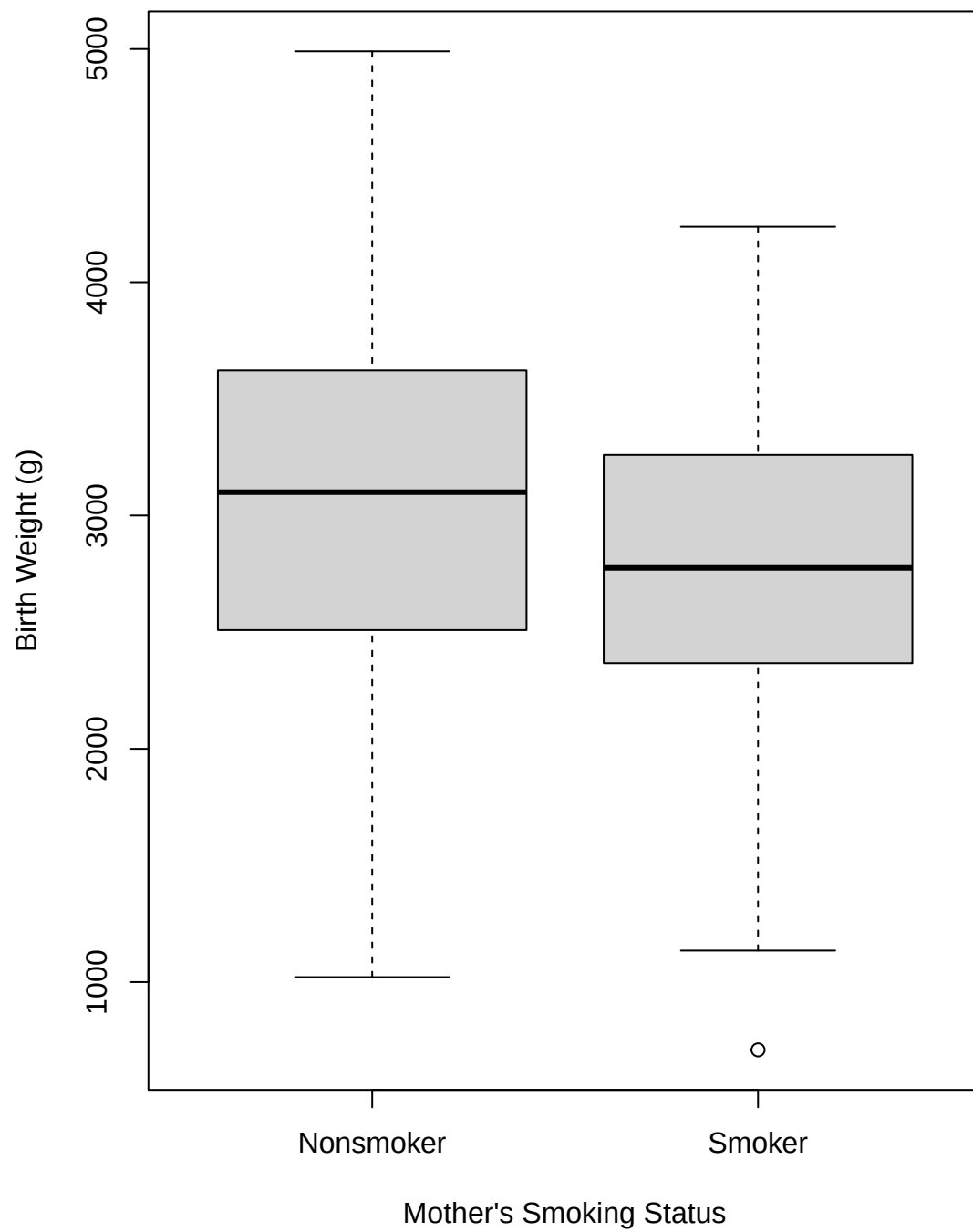


Figure 3.1: Side-by-side boxplots of birth weight based on mother's smoking status.

💡 Answer

From the boxplot it looks like the average baby weight is higher for nonsmoking mothers, however there is a large overlap in the IQR. The lower end of the boxplots look similar. I would be interesting in learning how many participants were smokers/nonsmokers, which we cannot tell from the boxplots.

! Association does not equal causation!

There may be a **lurking variable** which affects both the explanatory and response variable. This is why you should be careful about drawing conclusions just from statistical data.

Good study design should help eliminate potential lurking variables, however they are impossible to completely eliminate.

Example 3.3. What are potential lurking variables that could impact the study in Example 3.2?

Example 3.4. [The above data is available for you to manipulate on StatCrunch here.](#) Use the data to answer the following questions:

1. How many mothers were smokers/nonsmokers?
2. Is there a difference between the mother's weight for smokers/nonsmokers?
3. Is there a difference in the mother's age for smokers/nonsmokers?

Another method of visualizing relationships between a categorical explanatory variable and a quantitative response variable is by using a stacked histogram.

Here we see that the histogram is colored by player position. From this visualization we can see that defenders typically don't score much.

StatCrunch will let you do something similar, although it won't let you exactly stack the bars like above. In StatCrunch you can choose Graph > Histogram. Then you can select your response variable and group by your explanatory variable. Scroll all the way down to the multiple graph section and select the same X and Y axis. You can also use the rows and columns per page to change how many histograms are viewed at once.

3.2 Contingency Tables ($C \rightarrow C$)

Example 3.5. A college polls 200 students to determine whether there is a relationship between a student's class year (freshman, sophomore, junior, or senior) and whether or not they live off campus. This data is shown in the **contingency table** below.

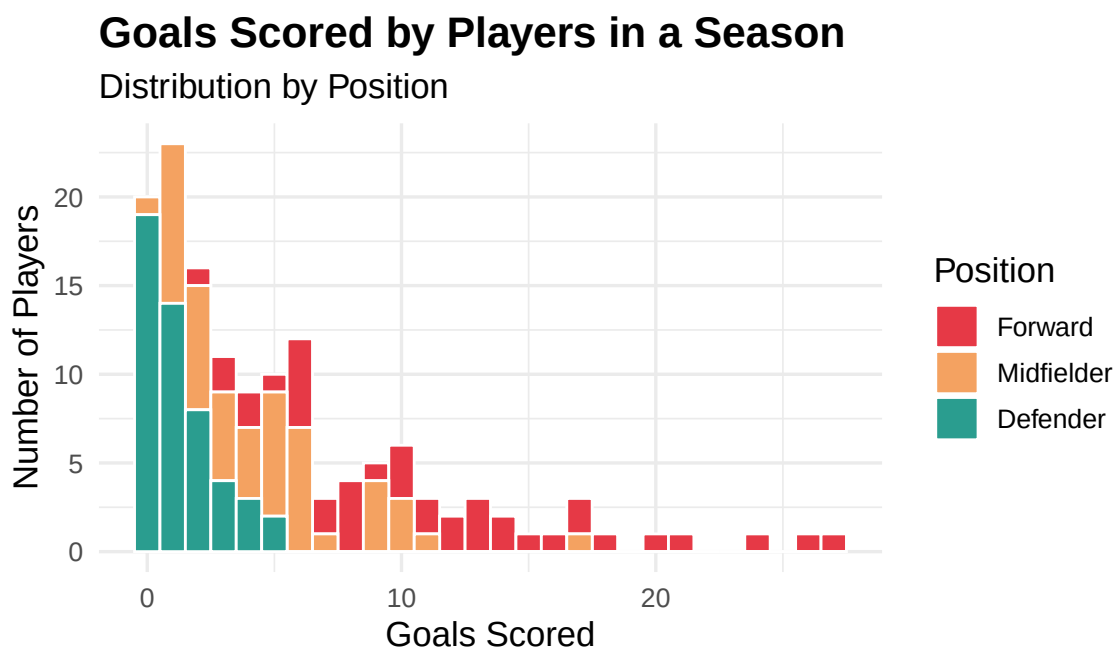


Figure 3.2: Stacked histogram of goals scored by different positions.

- What percentage of freshmen live off campus?
- What percentage of seniors live off campus?

Class Year	On Campus	Off Campus	Total
Freshman	42	13	55
Sophomore	35	15	50
Junior	18	30	48
Senior	9	38	47
Total	104	96	200

💡 Answer

- $(13/55) \times 100 = 23.6\%$
- $(38/47) \times 100 = 80.9\%$

We can convert a contingency table to percents. To do this we convert counts to percentages when restricting to a single explanatory variable. For example the row

Class Year	On Campus	Off Campus	Total
Freshman	42	13	55

becomes

Class Year	On Campus	Off Campus	Total
Freshman	$(42/55) \times 100 = \mathbf{76.4\%}$	$(13/55) \times 100 = \mathbf{23.6\%}$	100%

In this case the rows should each add up to 100%. If our explanatory variables were in the columns, then we would make the columns sum to 100%.

Example 3.6. Using the data from the baby weight study above, create a contingency table on StatCrunch of the mother’s smoking status and race. [Data](#)

1. For the mother’s at the hospital, does race impact whether or not they smoke? Use data from the table to support your answer.
2. Does smoking impact whether the mother’s are diagnosed with hypertension?

3.3 Scatterplots (Q -> Q)

Example 3.7. We want to determine if more time spent studying improves the score received on an exam. The amount of time spent studying was self reported by students. [Open the data in StatCrunch.](#)

Since we want to know if study time improves exam score, the explanatory variable is study time and the response variable is exam score. We can plot these using a scatter plot (Graph > Scatter Plot). We should set the study time as the x variable and the exam score as the y variable.

We see from the scatter plot that hours spent studying is **positively correlated** with exam scores. This means that the more hours spent studying, the higher of an exam score a student tends to receive. Notice that the points appear to show a “fuzzy” line with a positive slope.

Average hours of exercise weekly is **negatively correlated** with resting heart rate. This means the more hours exercised, the lower the resting heart rate.

We use the words **strong**, **moderate** and **weak** to describe how linearly correlated the data is. Below is an example of each,

Above the value r is the **correlation coefficient** for the data. Don’t worry about the formula for computing r as we will use StatCrunch for this computation. See how to use it below,

Example 3.8 (Correlation Coefficient). [Open this data in StatCrunch.](#) Make a scatter plot with longevity as the explanatory variable and gestation period as the response variable. Then compute the correlation coefficient of the data.

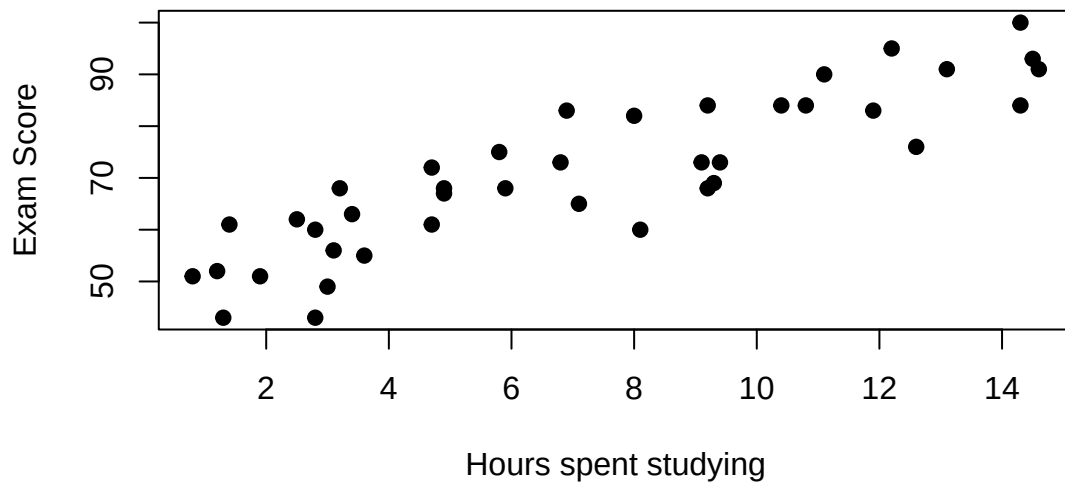


Figure 3.3: Scatter plot of time spent studying and exam scores.

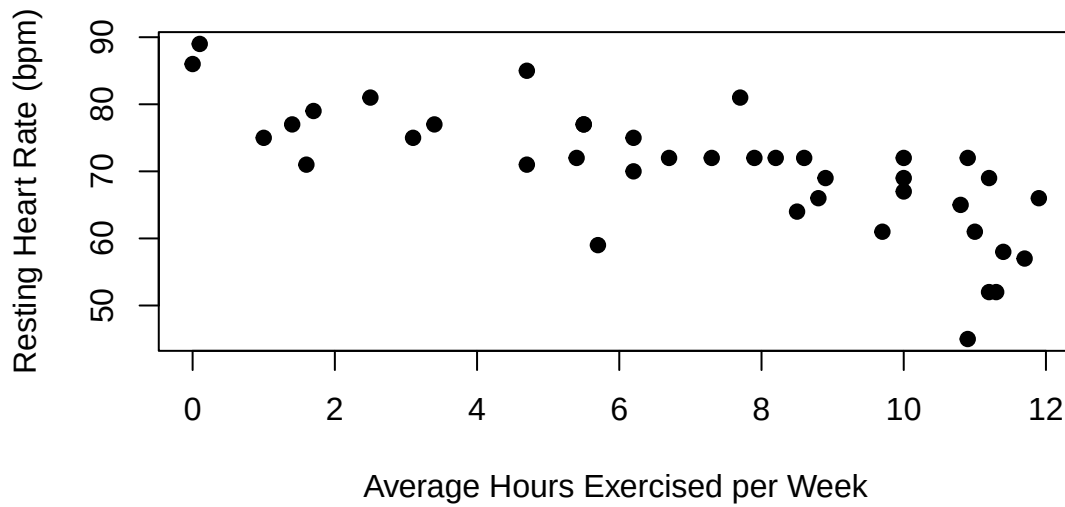


Figure 3.4: Scatter plot of weekly exercise hours and resting heart rate.

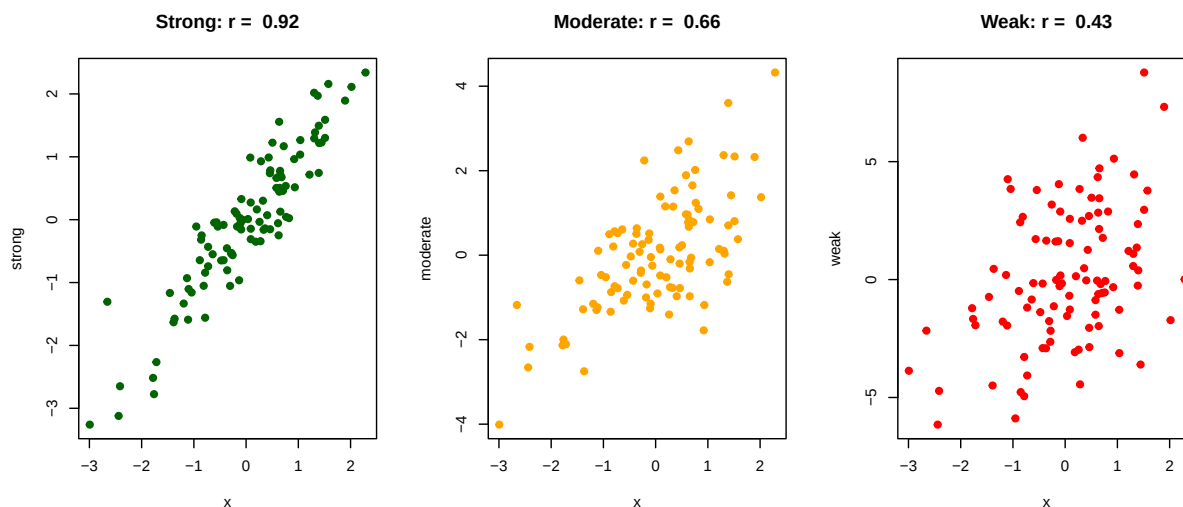


Figure 3.5: Strength of correlation.

The correlation coefficient r can measure the strength of the correlation, but only after we use the scatterplot to justify that it is linear! If our data is approximately linear, then the sign of the correlation coefficient tells us whether it is positive or negatively correlated. The magnitude of the correlation coefficient tells us the strength. There are a lot of different benchmarks you will see for what r value is “moderate” versus “strong.” You should think of these benchmarks as a guideline and that they are context dependent.

Plenty of data is not linearly correlated! Let’s see some examples.

! Don’t Over-Value r

The correlation coefficient should only be used as *supporting evidence* alongside the scatter plot. When a scatter plot looks linear, then r can tell us the strength and direction of that linearity. We cannot use r alone to justify linearity!!

3.4 Linear Regression

If we do feel a line is linear we can perform **least-squares linear regression**. This creates a line that best fits the data. We will not do this by hand, we will use StatCrunch. It is important to be able to interpret the StatCrunch output though.

Example 3.9. [Open this data in StatCrunch.](#) Perform the linear regression on the data. What is the equation for the line of best fit? Write a sentence to interpret the slope in the context of the explanatory and response variables.

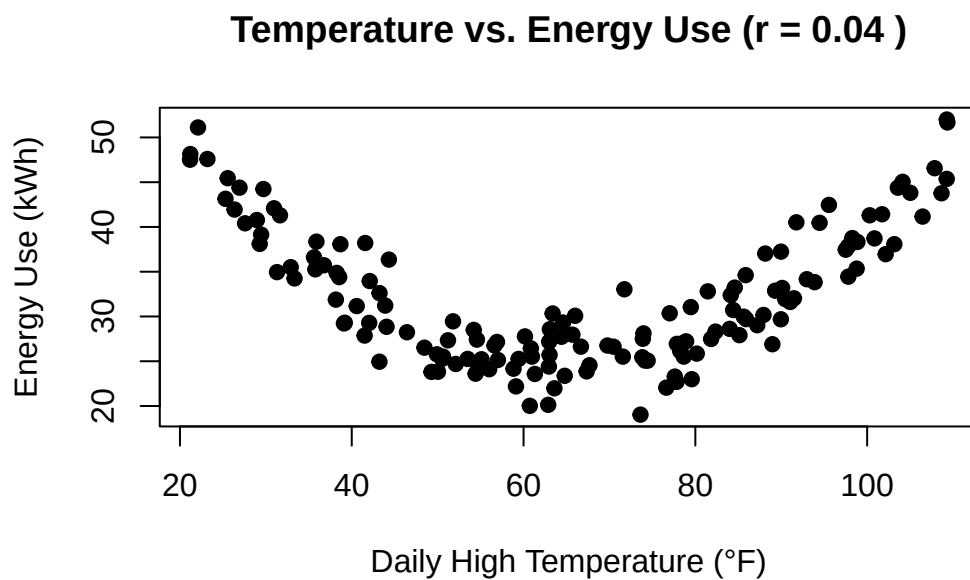


Figure 3.6: Daily home energy use is highest on cold and hot days and lowest at mild temperatures.

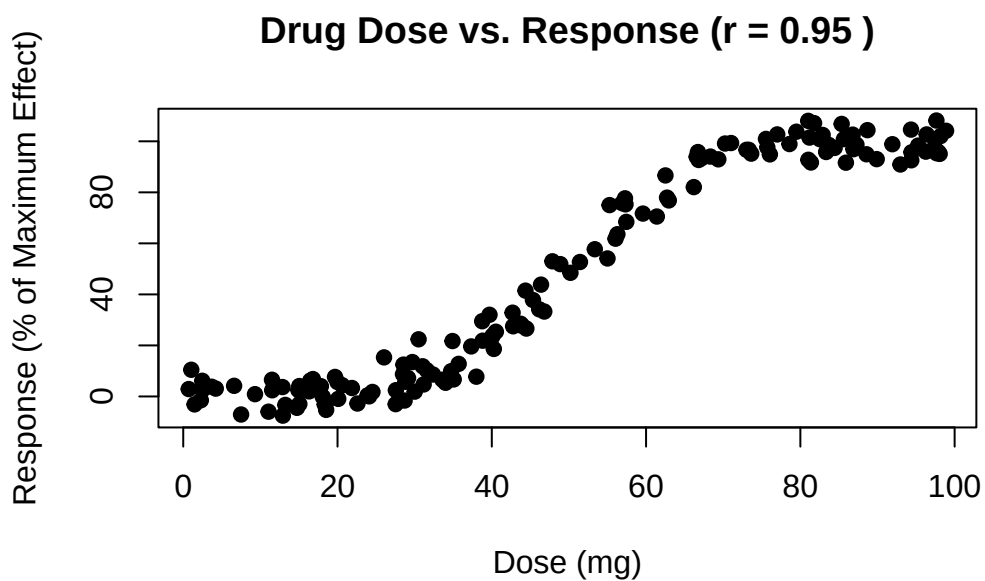


Figure 3.7: Response stays low at small doses, rises steeply, then plateaus near the maximum effect.

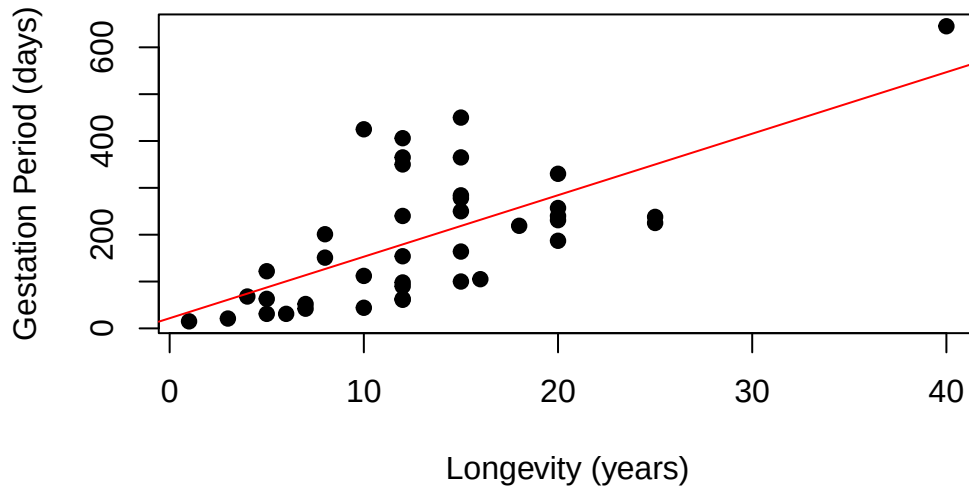


Figure 3.8: Longevity and gestation period for a variety of animals.

💡 Answer

Under the table **Parameter Estimates** we want to look at the **intercept** and **slope** estimate values to generate the line of best fit.

Intercept: 21.71

Slope: 13.13

Recall that longevity is our response variable—this means for our line it is the *input* (x variable). Gestation will be the *output* (y variable). So our line has the form,

$$\text{gestation} = 13.13 \times (\text{longevity}) + 21.71$$

The slope tells us that, on average, each year of longevity amounts to an additional 13.13 days of gestation. The intercept tells us for a animal with a longevity of 0 years, the gestation period is, on average, 21.71 days. This is an impossible scenario though, so it is outside of our domain of interest.

Example 3.10. Use the line of best fit computed in Example Example 3.9 to approximate the gestation period of a North American river otter, which has a typical lifespan of 12 years.

💡 Answer

Using our line, we estimate that the North American river otter will have a gestation period of $13.13 \times 12 + 21.71 = 179.27$ months.

According the the internet, the river otters have a gestation period of 285-375 days due to a process called delayed implantation. The more you know, eh?!

! Estimation

You should avoid using linear regression to estimate values that fall outside of the range of your data. When we zoom in on functions they almost always look linear, but zooming out can reveal more complicated behavior. Just because a line fits the data you have, doesn't mean it will still look linear when we "zoom out."

For example, the line in Example 3.9 should *not* be used to approximate the gestation time of an ORca, which averages 90 years of life. Plugging this in gives an approximate gestation period of 1,203 days or ~40 months—much longer than their actual gestation period of 15-18 months.

Example 3.11. Data. Using the provided dataset, answer the following:

1. Describe the relationship in words between the car's weight and its miles per gallon. Use the weight as the explanatory variable.
2. Compute the correlation coefficient.
3. Use linear regression to write a line that estimates the car's mpg from its weight.
4. Estimate the mpg of a car that weighs 4250 lbs.